

The objective of this paper is to provide a comparison and contrast of the code written by me and the code written by the AI tool of my choosing. The AI tool I choose as GitHub's CoPilot. I have used this tool in the past and use it on occasion for my internship. I will be focusing on the quality, the time, approaches, style, and things I would have done differently if done again.

When it comes down to the quality of the output in my code compared to the AI code I took a more comprehensive approach. I choose to add more print statements to include more frequent updates for the user. Although with that the console became a little bit harder to follow due to inconsistencies in line lengths. With the AI, it provided less print statements and had consistent line lengths making it easier to read the console's output. One thing that the AI did with the output to make it easier to comprehend what was going on was it gave the ability to assign players names. This made it easier to understand who was taking a turn.

Analyzing the time it took between the two, AI was able to get it done faster than me by roughly an hour. There are a few factors that I think that help contribute to that. The first major factor is sheer typing speed. I am by no means a fast typer or accurate typer. Thus it takes me longer to physically write the code out, whereas the AI can simply just generate it at a rapid pace. Another factor is debugging time, due to the simplicity of the assignment and the large library the AI has access to it was able to find a better solution than me faster. I had to take time relearning a bit of Java due to the fact that I have not coding in Java in over a year. However, as time progresses I believe that the time between the two would shorten.

Overall the AI did not need a ton of my assistance. The only place I really had to force myself to do something was when I was implementing my unique player into the AI's code. I gave it a few prompts asking it to create a new class for it and use my code but it either kept using its own code for that class or just duplicating the code it wrote for its unique player. This resulted in my having to copy and paste my own unique player class code.

The quality of the code itself varied. The AI had some weird formatting for some of the code that just seemed a little unnecessary or redundant. However, that may just come down to my personal preference

of how code should appear. When it came down to the debugging process. I largely spent time on figuring out a scanner issue where it just would refuse to work. I had to rearrange some code in order to get that fixed. With the AI it worked fine the first try. However that has not been my case with AI when I have used it in the past. With the AI written code as well I found it harder to debug because I had to take the time to understand what the AI had actually written. Whereas in my code it was easier because I knew what I had written and understood my personal thought process through it.

When comparing the style of the AI's code to mine. I definitely preferred mine over the AI's. I found that I was able to have it formatted in a way that was easier to understand for me. The organization was easy for me to follow and understand where things were. With the AI, it tended to put some code up at the top and "out of order." For example when writing the Prog6 class, it puts all the code in the public static void main(String[] args) part. I tend to like to have less "actual" code in that section for readability, understanding, and help with debugging.

If I were to do this project again, I would spend more time with the AI to try to get it to style and format the code in a way that is more understandable to me personally. I also allowed it to reference other files, I wonder if I had not given it access to reference those how much of a difference it would have made. If I were to hand do this project again with writing my own code, I would review your feedback and implement it as well as provide better documentation.